Milestone #4: Annotated Bibliography

Ricky Sparks
Research Methods - 693

February 10, 2026

Research Topic

Improving Uncertainty Communication in Scientific Visualization: A comparative study of error bars, distributional summaries (box/violin plots), and hypothetical outcome plots (HOPs) for decision accuracy and confidence calibration.

---

1. K. Mehta, "Maybe, Maybe Not: A Survey on Uncertainty in Visualization," arXiv preprint arXiv:2301.07687, 2022. [Online]. Available: https://arxiv.org/abs/2301.07687

   Annotation: Mehta surveys the importance of representing uncertainty in visualizations and the challenges authors and viewers face. The paper divides the visualization pipeline into four parts—data collection, preprocessing, visualization, and inference—and evaluates how uncertainty impacts each. It also explores authors' methodologies for processing and designing uncertainty and identifies future research directions. **Credibility**: arXiv preprints are not peer-reviewed, but the work is well-structured and cites established literature. The content is **accurate** in its characterization of the pipeline and challenges. The scope is **reasonable** for a survey. **Support** is limited to the author's synthesis and references; there is no original empirical study. The survey frames where uncertainty enters the pipeline and why it matters, reinforcing the motivation for studying how different encodings affect viewer interpretation.

2. B. Petek, D. Nabergoj, and E. Štrumbelj, "A General Approach to Visualizing Uncertainty in Statistical Graphics," arXiv preprint arXiv:2508.00937, 2025. [Online]. Available: https://arxiv.org/abs/2508.00937

   Annotation: Petek et al. propose a general method for visualizing uncertainty in static 2-D statistical graphics by treating a visualization as a function of underlying quantities;

uncertainty induces a distribution over images, which can be aggregated into a single display. The approach generalizes sample-based overlay methods, and standard representations such as confidence intervals emerge with coverage guarantees. The authors provide an open-source Python implementation and demonstrate both familiar and novel uncertainty visualizations. **Credibility**: arXiv preprint (not yet peer-reviewed), but the paper includes a formal treatment and implementation. The method is **accurate** and mathematically grounded. The claim that the approach is general is **reasonable** given the examples. **Support** comes from the implementation and examples. The paper adds another angle on encoding uncertainty in graphics and could inform the design of distributional summaries or HOPs-style encodings in the proposed study.

3. C. Stokes, C. Sanker, B. Cogley, and V. Setlur, "From Delays to Densities: Exploring Data Uncertainty through Speech, Text, and Visualization," arXiv preprint arXiv:2404.02317, 2024. [Online]. Available: https://arxiv.org/abs/2404.02317

   Annotation: Stokes et al. explore how speech, text, and visualization modes communicate data uncertainty and affect decision-making. They ran two crowdsourced experiments comparing variants within each mode. Visualization and text were most effective for rational decision-making, though text led to lower confidence; speech garnered the highest trust but sometimes led to riskier decisions. Stokes et al. highlight trade-offs among modes and call for more work on multimodal representations. **Credibility**: arXiv preprint; the study design is clear and the work addresses uncertainty communication empirically. The findings are **accurate** and grounded in empirical data. The scope is **reasonable**—focused on mode comparison rather than encoding comparison. **Support** comes from two crowdsourced experiments. The paper shows that visualization choice affects both decision quality and confidence, matching the proposed metrics of accuracy and confidence calibration.

4. S. Saklani, C. Goel, S. Bansal, Z. Wang, S. Dutta, T. M. Athawale, D. Pugmire, and C. R. Johnson, "Uncertainty-Informed Volume Visualization using Implicit Neural Representation," arXiv preprint arXiv:2408.06018, 2024. [Online]. Available: https://arxiv.org/abs/2408.06018

   Annotation: Saklani et al. propose uncertainty-aware implicit neural representations for volume visualization of scalar field datasets. They evaluate Deep Ensemble and Monte Carlo Dropout for uncertainty estimation and show that integrating prediction uncertainty enhances trustworthiness of DNN-based visualization. Saklani et al. show that uncertainty-informed volume visualization produces more informative and reliable results for scientific

data. **Credibility**: arXiv preprint; authors include Chris R. Johnson (SCI Institute, University of Utah) and Oak Ridge National Laboratory researchers. The method is **accurate** and evaluated across multiple datasets. The focus on scientific visualization is **reasonable** and aligned with the project domain. **Support** comes from extensive experiments and implementation. The paper situates the project in scientific visualization contexts where uncertainty communication matters for domain scientists.

5. J. Li, T. A. J. Ouermi, M. Han, and C. R. Johnson, "Uncertainty Tube Visualization of Particle Trajectories," arXiv preprint arXiv:2508.13505, 2025. [Online]. Available: https://arxiv.org/abs/2508.13505

   Annotation: Li et al. introduce the uncertainty tube, a visualization method for representing uncertainty in neural network-derived particle paths. The superelliptical tube captures nonsymmetric uncertainty and integrates techniques such as Deep Ensembles, Monte Carlo Dropout, and SWAG. The authors demonstrate the method on synthetic and simulation datasets. **Credibility**: arXiv preprint; authors are from the University of Utah (Chris R. Johnson). The design is **accurate** and computationally efficient. The approach is **reasonable** for trajectory data. **Support** comes from implementation and examples. The paper illustrates how novel uncertainty encodings can be designed for specific data types, tying into the project's goal of comparing encodings for scientific visualization tasks.

Citation style: IEEE. Annotations address Credibility, Accuracy, Reasonableness, and Support (CARS) and relevance to the research project.